# Marco Martinez

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## EXPERIENCE

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### Senior Software Engineer, FUJIFILM Healthcare Americas Corporation

**Research & Development** — Aug. 2026 - Present

*Selected to join a newly formed R&D team focused on shared infrastructure and modernization.*

- Modernized legacy per-service file logging by designing and implementing a centralized observability platform, providing a multi-target NuGet package for shared telemetry instrumentation and routing telemetry through an OpenTelemetry Collector deployed to Kubernetes.
- Deployed and configured Loki and Tempo with S3-compatible object storage for centralized logs and distributed traces, alongside Prometheus for time-series metrics.

*Technologies: C#, .NET 10, ASP.NET Core, .NET Standard 2.0, Kubernetes, OpenTelemetry*

### Product Development — Mar. 2025 - Aug. 2026

- Designed and delivered a third-party application integration POC that contributed to a successful $10M customer sale, implementing cross-window messaging to receive external events and persist corresponding updates to application data.
- Built a shared UI library standardizing dialog behavior and styling across 5+ web applications, securing cross-team adoption through technical review with engineering leads.
- Redesigned an enterprise chat application around participant-based conversation identity, preventing duplicate chats and automatically resuming existing conversations while modernizing the UI and interaction model.
- Optimized chat data retrieval by moving filtering and sorting into database procedures and limiting results, reducing response times from 6 seconds to near-instant in high-load environments.
- Diagnosed and resolved a production memory leak caused by unbounded event-listener registration that went undetected for several years, prompting a hotfix due to its business impact.

*Technologies: C#, .NET 10, .NET Framework 4.8, TypeScript, Razor Pages, PL/SQL*

### Software Engineer II, LexisNexis — May 2023 - Mar. 2025

- Designed data access and API-key authentication for an OData reporting integration consumed by Power BI, including database scripts, credential-management endpoints, and supporting services.
- Served as the primary technical point of contact for authentication integrations, guiding application and automation teams through API-key and token-based authentication flows.
- Led offshore engineers implementing OData endpoints, assigning work around repeatable application patterns and walking them through implementation conventions.
- Built a key event relationship framework for GraphQL request telemetry, enabling improved post-deployment troubleshooting while obfuscating PII values.
- Implemented an in-application query builder for dynamic data retrieval workflows.

*Technologies: C#, .NET 8, OData, GraphQL, Azure, T-SQL*

### **Software Engineer I, LexisNexis** — Feb. 2022 - May 2023

- Migrated multiple applications from implicit-flow authentication to PKCE code flow.
- Developed logout functionality for a shared authentication server used by multiple applications.
- Implemented custom CORS and host-filtering middleware to improve security for load-balanced cloud environments.
- Exposed OData endpoints and optimized GraphQL data retrieval workflows using data loaders.
- Implemented legacy Java override configurations to support cloud deployments.

*Technologies: C#, .NET 8, Java, OData, GraphQL, Azure*

### **Systems Engineer, Infosys** — Nov. 2021 - Feb. 2022

*Client: Walmart*

- Fixed bugs across enterprise applications using Java, Spring Boot, Kubernetes, and PL/SQL.
- Applied object-oriented design patterns and established application conventions while implementing Spring Boot services.

*Technologies: Java, Spring Boot, PL/SQL*

## PROJECTS

### Yavria — AI Context & Knowledge Platform

*C#, .NET, PostgreSQL, pgvector, React/TypeScript, Docker, OpenAI, OpenTelemetry*

- Built a self-hosted, multi-service AI platform for persistent conversation context, semantic knowledge retrieval, file-based context, and external application integrations.
- Designed a retrieval architecture using PostgreSQL/pgvector, provider-agnostic embeddings, streaming LLM responses, and explicit service boundaries between conversation orchestration and knowledge storage.
- Containerized and deployed the platform with ingress routing through Cloudflare Tunnel, centralized authentication and API access, database migrations, and diagnostics.

### NextProblem — Adaptive Coding Interview Practice Platform

*C#, ASP.NET Core, React/TypeScript, AI/LLM integration*

- Built a full-stack coding-practice platform that tracks problem history and generates personalized daily practice queues based on mastery, completion state, and spaced repetition.
- Designed a focus-first planning system that concentrates practice on individual problem-solving patterns while periodically resurfacing previously learned material.
- Integrated Yavria as an external AI coaching service, supplying practice history and progress as context for personalized guidance while keeping the two systems independently deployable.

## EDUCATION

**B.S. Computer Science** | Southern New Hampshire University